PRACTICE SHEET 1

Coulomb's Law & Electrostatic Force • Coulomb's Law & Charge Configurations • Electric Field due to Charge Distributions • Electric Field & Field Lines

Time Allowed: 60 Mins **Total Questions: 20**

Name: _____ Class/Sec: _____ Date: _____

Instructions: Attempt all questions. Each question carries one correct option unless marked *one or more than one option is correct*. Questions without options are numerical-value type: write the answer in the blank. The tag after each question shows its source examination. A self-assessment page follows; the answer key is provided as a separate sheet.

Q.1 In hydrogen like system the ratio of coulombian force and gravitational force between an electron and a proton is in the order of: *[JEE Main 2024, 5 Apr, Mor.]*

(A) 10^{19} (B) 10^{39} (C) 10^{29} (D) 10^{36}

Q.2 Force between two point charges q_1 and q_2 placed in vacuum at r cm apart is F. Force between them when placed in a medium having dielectric constant $K = 5$ at $r/5$ cm apart will be: *[JEE Main 2024, 31 Jan, Eve.]*

(A) $5F$ (B) $25F$ (C) $F/5$ (D) $F/25$

Q.3 If two charges q_1 and q_2 are separated with distance d and placed in a medium of dielectric constant K. What will be the equivalent distance between charges in air for the same electrostatic force? *[JEE Main 2023, 24 Jan, Mor.]*

(A) $d\sqrt{K}$ (B) $1.5\,d\sqrt{K}$

(C) $K\sqrt{d}$ (D) $2d\sqrt{K}$

Q.4 A charge of $4\,\mu$C is to be divided into two. The distance between the two divided charges is constant. The magnitude of the divided charges so that the force between them is maximum, will be: *[JEE Main 2022, 27 Jul, Eve.]*

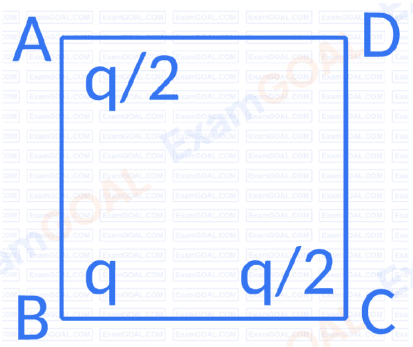
(A) $1\,\mu$C and $3\,\mu$C (B) $2\,\mu$C and $2\,\mu$C

(C) 0 and $4\,\mu$C (D) $1.5\,\mu$C and $2.5\,\mu$C

Q.5 Three charges $+Q$, q, $+Q$ are placed respectively, at distance 0, $d/2$ and d from the origin, on the x-axis. If the net force experienced by $+Q$, placed at $x = 0$, is zero, then value of q is: *[JEE Main 2019, 9 Jan, Mor.]*

(A) $-\dfrac{Q}{4}$ (B) $+\dfrac{Q}{2}$ (C) $+\dfrac{Q}{4}$ (D) $-\dfrac{Q}{2}$

Q.6 The three charges $q/2$, q and $q/2$ are placed at the corners A, B and C of a square of side a as shown in figure. The magnitude of electric field (E) at the corner D of the square, is: *[JEE Main 2022, 28 Jun, Mor.]*



(A) $\dfrac{q}{4\pi\epsilon_0 a^2}\left(\dfrac{1}{\sqrt{2}}+\dfrac{1}{2}\right)$

(B) $\dfrac{q}{4\pi\epsilon_0 a^2}\left(1+\dfrac{1}{\sqrt{2}}\right)$

(C) $\dfrac{q}{4\pi\epsilon_0 a^2}\left(1-\dfrac{1}{\sqrt{2}}\right)$

(D) $\dfrac{q}{4\pi\epsilon_0 a^2}\left(\dfrac{1}{\sqrt{2}}-\dfrac{1}{2}\right)$

Q.7 Ten charges are placed on the circumference of a circle of radius R with constant angular separation between successive charges. Alternate charges 1, 3, 5, 7, 9 have charge $(+q)$ each, while 2, 4, 6, 8, 10 have charge $(-q)$ each. The potential V and the electric field E at the centre of the circle are respectively. (Take $V = 0$ at infinity) *[JEE Main 2020, 5 Sep, Eve.]*

(A) $V = 0;\ E = 0$

(B) $V = \dfrac{10q}{4\pi\epsilon_0 R};\ E = \dfrac{10q}{4\pi\epsilon_0 R^2}$

(C) $V = \dfrac{10q}{4\pi\epsilon_0 R}$; $E = 0$

(D) $V = 0$; $E = \dfrac{10q}{4\pi\epsilon_0 R^2}$

Q.8 Four point charges $-q$, $+q$, $+q$ and $-q$ are placed on y-axis at $y = -2d$, $y = -d$, $y = +d$ and $y = +2d$, respectively. The magnitude of the electric field E at a point on the x-axis at $x = D$, with $D \gg d$, will behave as: *[JEE Main 2019, 9 Apr, Eve.]*

(A) $E \propto \dfrac{1}{D^3}$ (B) $E \propto \dfrac{1}{D}$

(C) $E \propto \dfrac{1}{D^4}$ (D) $E \propto \dfrac{1}{D^2}$

Q.9 Four charges equal to $-Q$ are placed at the four corners of a square and a charge q is at its center. If the system is in equilibrium the value of q is *[AIEEE 2004]*

(A) $-\dfrac{Q}{2}\left(1+2\sqrt{2}\right)$ (B) $\dfrac{Q}{4}\left(1+2\sqrt{2}\right)$

(C) $-\dfrac{Q}{4}\left(1+2\sqrt{2}\right)$ (D) $\dfrac{Q}{2}\left(1+2\sqrt{2}\right)$

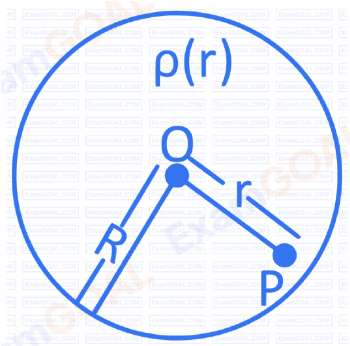
Q.10 A thin half ring of radius 35 cm is uniformly charged with a total charge of Q coulomb. If the magnitude of the electric field at centre of the half ring is 100 V/m, then the value of Q is _____ nC. ($\epsilon_0 = 8.85 \times 10^{-12}$ C^2/Nm^2 and $\pi = 3.14$) *[JEE Main 2026, 6 Apr, Mor.]*

(A) 2.14 (B) 2.44 (C) 3.25 (D) 0.7

Q.11 σ is the uniform surface charge density of a thin spherical shell of radius R. The electric field at any point on the surface of the spherical shell is: *[JEE Main 2024, 6 Apr, Mor.]*

(A) $\sigma/\epsilon_0 R$ (B) σ/ϵ_0 (C) $\sigma/2\epsilon_0$ (D) $\sigma/4\epsilon_0$

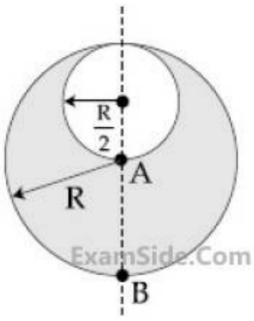
Q.12 A spherically symmetric charge distribution is considered with charge density varying as $\rho(r) = \rho_0\left(\dfrac{3}{4} - \dfrac{r}{R}\right)$ for $r \le R$ and $\rho(r) = 0$ for $r > R$. Here r ($r < R$) is the distance from the centre O (as shown in figure). The electric field at point P will be: *[JEE Main 2022, 29 Jul, Mor.]*



(A) $\dfrac{\rho_0 r}{4\epsilon_0}\left(\dfrac{3}{4} - \dfrac{r}{R}\right)$ (B) $\dfrac{\rho_0 r}{3\epsilon_0}\left(\dfrac{3}{4} - \dfrac{r}{R}\right)$

(C) $\dfrac{\rho_0 r}{4\epsilon_0}\left(1 - \dfrac{r}{R}\right)$ (D) $\dfrac{\rho_0 r}{5\epsilon_0}\left(1 - \dfrac{r}{R}\right)$

Q.13 Consider a sphere of radius R which carries a uniform charge density ρ. If a sphere of radius $\frac{R}{2}$ is carved out of it, as shown, the ratio $\left|\dfrac{\vec{E}_A}{\vec{E}_B}\right|$ of magnitude of electric field $\vec{E}_A$ and $\vec{E}_B$, respectively, at points A and B due to the remaining portion is: *[JEE Main 2020, 9 Jan, Mor.]*



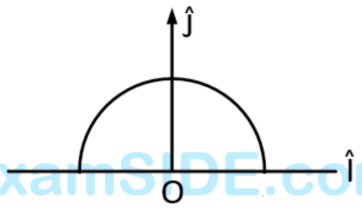
(A) $\dfrac{17}{54}$ (B) $\dfrac{18}{54}$

(C) $\dfrac{18}{34}$ (D) $\dfrac{21}{34}$

Q.14 For a uniformly charged ring of radius R, the electric field on its axis has the largest magnitude at a distance h from its center. Then value of h is: *[JEE Main 2019, 9 Jan, Mor.]*

(A) $\dfrac{R}{\sqrt{5}}$ (B) $\dfrac{R}{\sqrt{2}}$ (C) R (D) $R\sqrt{2}$

Q.15 A thin semi-circular ring of radius r has a positive charge q distributed uniformly over it. The net field $\vec{E}$ at the centre O is *[AIEEE 2010]*



(A) $\frac{q}{4\pi^2\epsilon_0 r^2}\hat{j}$ (B) $-\frac{q}{4\pi^2\epsilon_0 r^2}\hat{j}$
(C) $-\frac{q}{2\pi^2\epsilon_0 r^2}\hat{j}$ (D) $\frac{q}{2\pi^2\epsilon_0 r^2}\hat{j}$

Q.16 A point charge $+q$ is placed at the origin. A second point charge $+9q$ is placed at $(d, 0, 0)$ in Cartesian coordinate system. The point in between them where the electric field vanishes is: *[JEE Main 2025, 2 Apr, Mor.]*

(A) $(3d/4, 0, 0)$ (B) $(d/4, 0, 0)$
(C) $(4d/3, 0, 0)$ (D) $(d/3, 0, 0)$

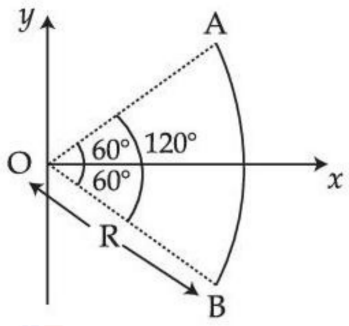
Q.17 Two charges q and $3q$ are separated by a distance r in air. At a distance x from charge q, the resultant electric field is zero. The value of x is: *[JEE Main 2024, 31 Jan, Mor.]*

(A) $\frac{r}{3(1+\sqrt{3})}$ (B) $\frac{(1+\sqrt{3})}{r}$
(C) $\frac{r}{(1+\sqrt{3})}$ (D) $r(1+\sqrt{3})$

Q.18 Two point charges A and B of magnitude $+8\times10^{-6}$ C and -8×10^{-6} C respectively are placed at a distance d apart. The electric field at the middle point O between the charges is 6.4×10^4 N C^{-1}. The distance d between the point charges A and B is: *[JEE Main 2022, 28 Jun, Eve.]*

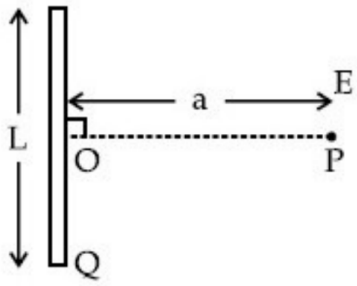
(A) 2.0 m (B) 3.0 m (C) 1.0 m (D) 4.0 m

Q.19 Figure shows a rod AB, which is bent in a 120° circular arc of radius R. A charge $(-Q)$ is uniformly distributed over rod AB. What is the electric field $\vec{E}$ at the centre of curvature O? *[JEE Main 2021, 27 Aug, Eve.]*



(A) $\frac{3\sqrt{3}Q}{8\pi\epsilon_0 R^2}(\hat{i})$ (B) $\frac{3\sqrt{3}Q}{8\pi^2\epsilon_0 R^2}(\hat{i})$
(C) $\frac{3\sqrt{3}Q}{16\pi^2\epsilon_0 R^2}(\hat{i})$ (D) $\frac{3\sqrt{3}Q}{8\pi^2\epsilon_0 R^2}(-\hat{i})$

Q.20 Find the electric field at point P (as shown in figure) on the perpendicular bisector of a uniformly charged thin wire of length L carrying a charge Q. The distance of the point P from the centre of the wire is $a = \frac{\sqrt{3}}{2}L$. *[JEE Main 2021, 26 Feb, Mor.]*



(A) $\frac{Q}{4\pi\epsilon_0 L^2}$ (B) $\frac{Q}{3\pi\epsilon_0 L^2}$
(C) $\frac{Q}{2\sqrt{3}\pi\epsilon_0 L^2}$ (D) $\frac{\sqrt{3}Q}{4\pi\epsilon_0 L^2}$

SELF-ASSESSMENT – PRACTICE SHEET 1

Time Allowed: 60 Mins	**Time Taken:** _____ Mins
Total Questions: 20	**Questions Attempted:** _____
Correct: _____	**Incorrect:** _____
Unattempted: _____	**Accuracy:** _____ % (Correct / Attempted ×100)

Question-wise record (tick one: C = Correct, W = Wrong, S = Skipped)

Q.1 ☐C ☐W ☐S	**Q.2** ☐C ☐W ☐S	**Q.3** ☐C ☐W ☐S	**Q.4** ☐C ☐W ☐S
Q.5 ☐C ☐W ☐S	**Q.6** ☐C ☐W ☐S	**Q.7** ☐C ☐W ☐S	**Q.8** ☐C ☐W ☐S
Q.9 ☐C ☐W ☐S	**Q.10** ☐C ☐W ☐S	**Q.11** ☐C ☐W ☐S	**Q.12** ☐C ☐W ☐S
Q.13 ☐C ☐W ☐S	**Q.14** ☐C ☐W ☐S	**Q.15** ☐C ☐W ☐S	**Q.16** ☐C ☐W ☐S
Q.17 ☐C ☐W ☐S	**Q.18** ☐C ☐W ☐S	**Q.19** ☐C ☐W ☐S	**Q.20** ☐C ☐W ☐S

Reason for mistakes (tick all that apply)

☐ Concept not clear ☐ Formula forgotten ☐ Calculation mistake
☐ Misread the question ☐ Ran out of time ☐ Guess work

Personal Feedback

What went well:

_____

_____

Topics / questions I need to revise:

_____

_____

My action plan before the next sheet:

_____

_____

Teacher's remarks:

_____

_____

Student's signature: _____ Teacher's signature: _____